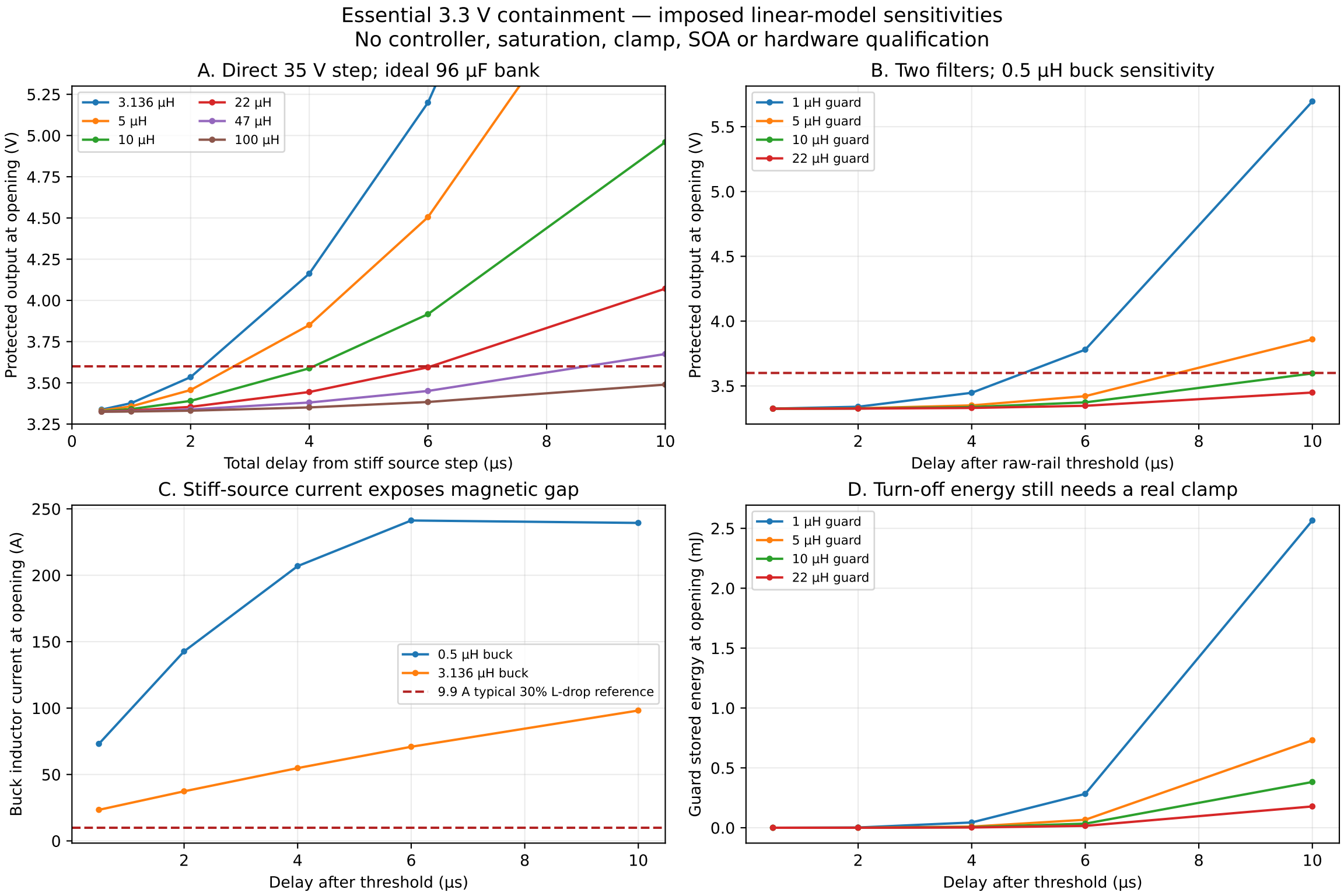
Essential 3.3 V containment — imposed linear-model sensitivities
No controller, saturation, clamp, SOA or hardware qualification
A. Direct 35 V step; ideal 96 µF bank
Protected output at opening (V)
Total delay from stiff source step (µs)
3.136 µH
5 µH
10 µH
22 µH
47 µH
100 µH
B. Two filters; 0.5 µH buck sensitivity
Protected output at opening (V)
Delay after raw-rail threshold (µs)
1 µH guard
5 µH guard
10 µH guard
22 µH guard
C. Stiff-source current exposes magnetic gap
Buck inductor current at opening (A)
Delay after threshold (µs)
0.5 µH buck
3.136 µH buck
9.9 A typical 30% L-drop reference
D. Turn-off energy still needs a real clamp
Guard stored energy at opening (mJ)
Delay after threshold (µs)
1 µH guard
5 µH guard
10 µH guard
22 µH guard